

# Ratio Notation and Sharing

GCSE Maths · Practice Questions

20 questions · 62 marks total

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## Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Write the ratio 15 : 25 in its simplest form.

.....

2 [2 marks]

Write the ratio 24 : 36 : 60 in its simplest form.

.....

3 [2 marks]

Share £45 in the ratio 4 : 5

.....

4 [2 marks]

Write the ratio 2 : 5 in the form 1 : n

.....

5 [3 marks]

120 sweets are shared between three children in the ratio 3 : 4 : 5

How many sweets does each child get?

.....

6

[3 marks]

In a box of counters the ratio of red to blue is 2 : 7

There are 14 red counters.

- (a) How many blue counters are there?  
(b) How many counters are there altogether?
- 

7

[3 marks]

Write the ratio 45 minutes : 2 hours in its simplest form.

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8

[3 marks]

A pastry mix uses flour and butter in the ratio 5 : 2

350 g of flour is used.

- (a) How much butter is needed?  
(b) What is the total mass of the mixture?
- 

9

[4 marks]

Some money is shared in the ratio 3 : 8

The larger share is £75 more than the smaller share.

Work out the total amount of money.

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10

[4 marks]

$a : b = 2 : 3$  and  $b : c = 4 : 5$

Find the ratio  $a : b : c$  in its simplest form.

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Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1 [2 marks]

Write the ratio  $0.4 : 1.2$  in its simplest form.

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2 [2 marks]

Write the ratio  $5 : 8$  in the form  $n : 1$

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3 [3 marks]

$x : y = 3 : 4$  and  $y : z = 6 : 7$

Find  $x : y : z$

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4 [3 marks]

In a class the ratio of boys to girls is  $5 : 7$

(a) What fraction of the class are boys?

(b) There are 21 girls. How many students are in the class?

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5 [4 marks]

A bag contains red and blue counters in the ratio  $3 : 5$

12 more red counters are added and the ratio becomes  $9 : 10$

Work out the total number of counters now in the bag.

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6

[4 marks]

A money box contains 20p coins and 50p coins in the ratio 5 : 3  
Six of the 50p coins are removed, and the ratio becomes 5 : 2  
How many coins were in the box originally?

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7

[4 marks]

A bag contains 10p, 20p and 50p coins in the ratio 4 : 5 : 6  
The total value of the coins is £8.80  
How many coins are there altogether?

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8

[4 marks]

Some money is shared between A, B and C in the ratio 2 : 3 : 7  
C receives £120 more than B.  
Work out how much each person receives.

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9

[4 marks]

Paint is mixed using white and blue in the ratio 7 : 3  
Tom needs 25 litres of the mixture but only has 6 litres of blue paint.  
(a) Show that Tom cannot make 25 litres of the mixture.  
(b) Work out the greatest volume of mixture he can make.

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A map has a scale of 1 : 50 000

A rectangular field measures 4 cm by 3 cm on the map.

(a) Work out the real dimensions of the field in kilometres.

(b) Work out the real area of the field in  $\text{km}^2$ .

(c) Explain why the area on the map is not  $\frac{1}{50\,000}$  of the real area.